

Hydroclimatological Perspective of the Kerala Flood of 2018

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ABSTRACT

Flood is among the deadliest disasters in India, and the frequency of floods and extreme precipitation events is projected to increase under the warming climate. The frequency of floods in India varies geographically as some regions are more prone to floods than the others. The Kerala flood of 2018 caused enormous economic damage, affected millions of people, and resulted in the death of more than 400 people. Here we provide a hydroclimatological perspective on the Kerala flood of 2018. Using the observations and model simulations from the Variable Infiltration Capacity (VIC) model, we show that the 2018 extreme precipitation and runoff conditions that caused flooding were unprecedented in the record of the past 66 years (1951-2017). Our results show that mean monsoon precipitation has significantly declined while air temperature has significantly increased during 1951-2017 in Kerala. The drying and warming trends during the monsoon season resulted in a declined total runoff in large part of the state in the last 66 years. Apart from the mean hydroclimatic conditions, extreme precipitation, and extreme total runoff have also declined from 1951 to 2017. However, 1 and 2-day extreme precipitation and extreme runoff conditions in August 2018 exceeded substantially from the long-term 95th percentiles recorded during 1951-2017. Since there is no increase in mean and extreme precipitation in Kerala over the last six decades, the extreme event during August 2018 is likely to be driven by anomalous atmospheric conditions due to climate variability rather anthropogenic climate warming. The severity of the Kerala flood of 2018 and the damage caused might be affected by several factors including land use/land cover change, antecedent hydrologic conditions, reservoir storage and operations, encroachment of flood plains, and other natural factors. The impacts of key drivers (anthropogenic and natural) on flood severity need to be established to improve our understanding of floods and associated damage.

INTRODUCTION

Extreme precipitation and floods are the hydroclimatic events that draw considerable attention every year during the monsoon season in India. Millions of people are affected by extreme precipitation and flood events that often cause damage to infrastructure and agriculture in India. Extreme precipitation events have increased in India during the last few decades, which are likely to increase further under the warming climate (Goswami et al., 2006; Mukherjee et al., 2018; Roxy et al., 2017). India witnessed some of the most severe extreme precipitation and flood events during the recent decades. For instance, the extreme precipitation event in Mumbai in 2005 affected more than 20 million people and caused more than 1000 deaths (Gupta and Nair, 2011). The 2013 extreme rainfall and flood event in Uttarakhand resulted in the death of more than 6000 people and caused an economic loss of more than \$3.8 billion (Kumar, 2013). Furthermore, extreme precipitation in Chennai in November 2015 resulted in the economic loss of over \$3 billion (Boyaj et al., 2018; Van Oldenborgh et al., 2016).

The extreme precipitation and associated flooding in Kerala in August 2018 had a remarkable impact on all the aspects of the social life of millions of people. Preliminary estimates suggest that an economic loss of more than \$3 billion and death of more than 440 people caused by the flood of 2018 in Kerala (Gulf News, 30 August 2018). Mishra et al. (2018) reported that the Kerala flood was caused by the above seasonal precipitation, multi-day extreme precipitation, above normal reservoir storage, and extreme precipitation in the catchments upstream of the reservoirs. One and 2-day extreme precipitation that occurred on 15-16th August 2018 in Kerala had a return period of about 75 and 200 years, respectively based on the long-term record (1901-2017) (Mishra et al., 2018). Here, we provide a hydroclimatological perspective on the Kerala flood of 2018. We evaluate the long-term (1951-2017) changes in mean and extreme precipitation, air temperature, and total runoff in the state of Kerala. Moreover, we evaluate the changes in 1 and 2-day annual maximum precipitation and total runoff in Kerala in the context of the 2018 flood. Finally, we compare the 2018 hydroclimatological conditions to the long-term (1851-2017) record to find out the anomalous nature of the 2018 event.

DATASET

Climate Data

We obtained daily gridded precipitation, maximum and minimum temperatures from India Meteorological Department (IMD) at 0.25 degree spatial resolution for the period 1951-2018 (till September 2018). IMD monitors daily precipitation at more than 6000 stations across India. IMD precipitation represents the key features of the Indian summer monsoon and also captures the variability in rainfall due to orography, which is an essential feature in Kerala (Pai et al., 2015; Rajeevan et al., 2006). We obtained maximum and minimum temperatures at 0.5° spatial resolution (Srivastava et al., 2009) from IMD, which were regridded to 0.25° using Synergic mapping (SYMAP) as described in (Maurer et al., 2002). The SYMAP method considers temperature lapse rate to account for variation in temperature with topography. We obtained NCEP-NCAR re-analysis daily wind speed data, which were regridded to 0.25 degree to make it consistent with precipitation and temperature. Gridded precipitation and temperature from IMD have been widely used for hydroclimatic studies in India (Mishra et al., 2018; Shah and Mishra, 2016b, 2016a; R.D. Shah and Mishra, 2015; R. Shah and Mishra, 2014a).

The Variable Infiltration Capacity (VIC) model

We used the VIC model to simulate hydrologic conditions during August 2018 Kerala flood. The VIC model (Cherkauer et al., 2003; Liang et al., 1996) is a macroscale, grid based, semi-distributed hydrological model, that simulates full water and energy budget at each grid-cell. The VIC model can represent the sub-grid variability of elevation, soil, and vegetation parameters. We obtained land use land cover data from National Remote Sensing Center (NRSC) at 56 m resolution while the other vegetation parameters (leaf area

index and albedo) were obtained from Moderate Resolution Spectroradiometer (MODIS). Soil parameters were obtained from the Harmonized World Soil Database (HWSD: <http://www.fao.org/soils-portal/soil-survey/soil-maps-and-databases/harmonized-world-soil-database-v12/en/>). The VIC model simulates surface runoff and baseflow for each grid cell, and we analyzed total runoff (surface runoff+baseflow) in Kerala to understand long-term (1951-2017) trends and variability. Surface runoff is defined by the variable infiltration curve (Wood et al., 1992), which is a non-linear function of the fractional saturated area. On the other hand, baseflow is estimated using non-linear Arno baseflow model (Franchini and Pacciani, 1991). We obtained the VIC model calibration parameters from H.L. Shah and Mishra, (2016a), therefore, the model was not explicitly calibrated for this study. The VIC model has been extensively used to study droughts and floods in India (Mishra et al., 2014; H.L. Shah and Mishra, 2016b). Moreover, Aadhar and Mishra (2017) developed a real-time drought monitoring system using the VIC model.

Analytical Approach

We analyzed mean monsoon (June to September) season precipitation, air temperature, and the VIC model simulated total runoff over Kerala state from 1951 to 2017. We estimated changes in mean monsoon season precipitation, air temperature, and simulated total runoff using the nonparametric Mann-Kendall (M-K) test and Sen's

slope method at 5% significance level. We estimated the trend in 1-day and 2-day maximum precipitation and total runoff from 1951 to 2017. The M-K test has been widely used to estimate trend and change in hydrologic and climate datasets (Mishra and Lettenmaier, 2011; H. L. Shah and Mishra, 2016a). 1-day and 2day maximum precipitation and total runoff for 2018 were compared with the long-term (1951-2017) mean, 75th and 95th percentiles.

RESULTS

Long-term Hydroclimatology of Kerala

Since we are interested to understand the hydroclimatological perspective of the Kerala flood of 2018, we first analyzed the long-term (1971-2017) mean and trends associated with the monsoon season precipitation, air temperature, and total runoff (Fig. 1). We found that there is a large variability in the monsoon season precipitation from northern to southern part of Kerala (Fig.1a). For instance, mean monsoon precipitation varied from 600-3000 mm in Kerala during the period of 1951-2017 (Fig. 1), whereas maximum precipitation occurs in North and central Kerala. Similarly, we noticed a large variability in the monsoon season mean air temperature during the period of 1951-2017. For instance, mean air temperature varied between 22 and 30° C in Kerala with a higher temperature in the southern part. We simulated total runoff, which is surrogate of

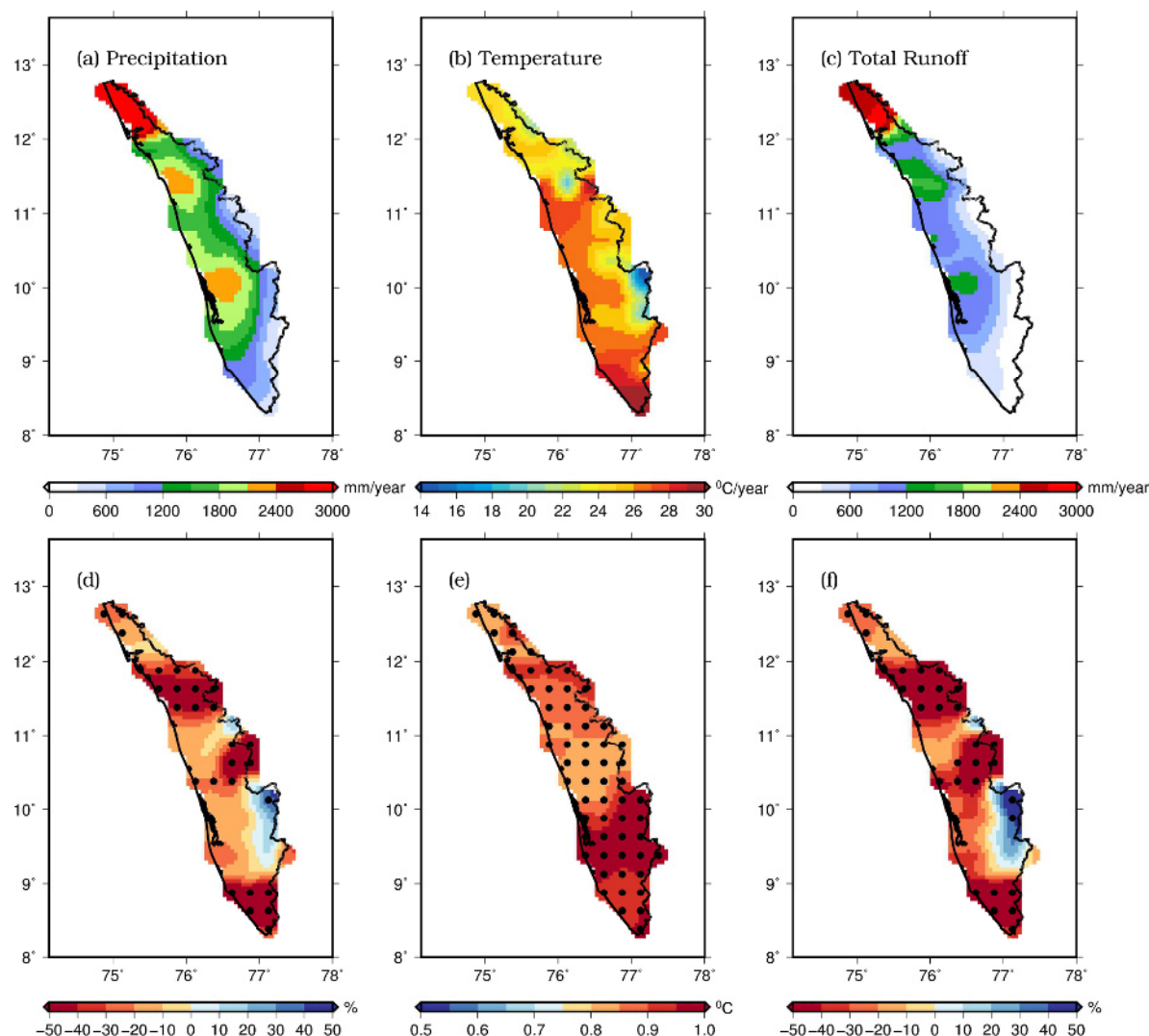


Fig.1. (a-c) Mean monsoon precipitation, temperature and total runoff from 1951-2017. **(d-f)** The trend in mean monsoon precipitation, air temperature, and total runoff during 1951-2017. The trend was estimated using the Mann-Kendall trend test and Sen's slope method. Stippling in (d-f) shows the statistically significant trend at 5% significance level.

streamflow, for the period of 1951-2017 using the VIC model. We found that simulated total runoff for the monsoon season exhibits a similar pattern to the monsoon season precipitation (Fig. 1c). For instance, total runoff in Kerala varies from 300 to 2500 mm with higher values in the northern parts while lower values in the southern regions of the state (Fig. 1c). Overall, the long-term climatological mean of precipitation, air temperature, and total runoff shows a large spatial variability in Kerala that is dominated by higher precipitation and runoff conditions in northern Kerala.

One important question that arises after the Kerala flood of August 2018 is whether the state had been witnessing significant changes in the monsoon season precipitation and temperature during the period of 1951-2017. Therefore, we estimated the changes in hydroclimatic conditions during the monsoon season in Kerala (Fig. 1d-f). We find a declining trend in mean monsoon precipitation during 1951-2017. We noticed the declining trend in the monsoon season precipitation in Kerala and that is evident across the state with a significant decline of more than 20% in the northern parts of the state during the period of 1951-2017 (Fig. 1d). We also find that there has been significant warming in the monsoon season air temperature across the state in the last 66 years (Fig. 1e). More remarkably, the southern part of Kerala has warmed more than 1.0°C during 1951-2017. Our findings are consistent with H. L. Shah and Mishra (2016a) who showed increasing (decreasing) trend of temperature (precipitation) over Kerala. The drying and warming during the monsoon season in Kerala resulted in a significant (p -value < 0.05) decline in total runoff in the majority of the state between 1951 and 2017 (Fig. 1f). Overall, we find that the state of Kerala has witnessed significant changes in the long-term hydroclimatology.

We find that there have been substantial changes in the long-term hydroclimatology in Kerala. Next, we analyze the changes in the extreme precipitation and runoff conditions during 1951-2017 (Fig. 2). To do so, we selected 1 and 2-day annual maximum precipitation and annual maximum total runoff during the monsoon season for the 1951-2017 period. We find that similar to mean monsoon season precipitation, 1 and 2-day annual maximum precipitation has declined in the majority of the state with a significant decline in the northern part (Fig. 2a,c). Similarly, we find an overall decline in 1 and 2-day annual maximum runoff in the majority of Kerala during 1951-2017 (Fig. 2b,d). However, the northern part of Kerala experienced a significant decline in 1 and 2-day annual maximum total runoff, which is attributed to a long-term decline in the monsoon season precipitation and increase in air temperature in the state. Our results indicate a long-term decline in the mean and extreme precipitation and runoff conditions in the state during 1951-2017.

Hydroclimatic Conditions During August 2018

Next, we analyzed hydroclimatic conditions, especially extreme precipitation, and total runoff during August 2018 (Fig. 3). We estimated 1 and 2-day extreme precipitation and total runoff in Kerala during August 2018 to understand the enormity of extreme hydroclimatic conditions in the state that caused wide-spread flooding. We found that 1-day maximum precipitation varied between 100 and 400 mm during August 2018 in Kerala, which is substantially higher than its long-term mean. We find that the highest precipitation in Kerala during August 2018 occurred on 15th August (Mishra et al., 2018). The two-day maximum precipitation was more than 400 mm in large part of the state and some regions experienced more than 600mm precipitation in August 2018 (Fig. 3c). The two-day maximum precipitation in Kerala occurred on 15 and 16th August that caused enormous flooding in the state (Mishra et al., 2018).

Similar to extreme precipitation, 1 and 2-day maximum total runoff varied greatly within the state during August 2018. For instance, 1-day total runoff varied between 50 and 200mm in the state, which was

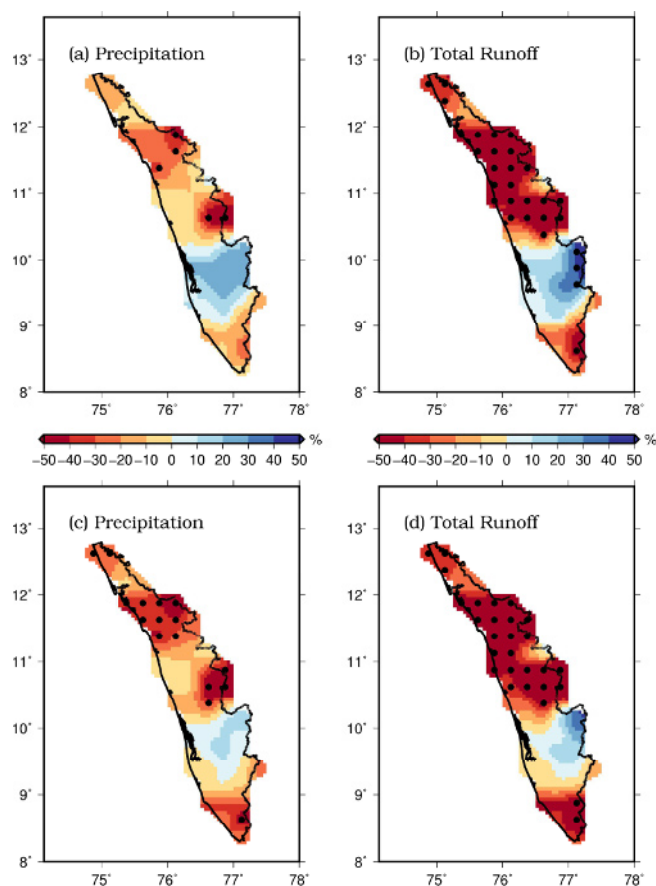


Fig.2. (a-b) Change (%) in 1-day annual maximum precipitation and total runoff for the period of 1951-2017. **(c-d)** Change (%) in 2-day annual maximum precipitation and total runoff from 1951-2017. The change was estimated using trend slope multiplied by the duration (1951-2017).

substantially higher than its long-term (1951-2017) climatology. Similarly, the 2-day maximum total runoff exceeded 250mm in large part of Kerala, which shows an extreme hydrologic condition that has contributed to the large-scale flooding in August 2018 (Fig. 3b, d). We calculated 1-day and 2-day maximum precipitation and total runoff for the 2018 year (Fig. 3).

We estimated mean annual monsoon precipitation, air temperature, and total runoff from 1951-2018. We find that mean monsoon precipitation (2000 mm) of 2018 was slightly below the 95th percentile of mean monsoon precipitation (2050 mm) estimated for the period of 1951-2015 (Fig. 4a). On the other hand, the monsoon season of 2018 was substantially cooler than its long-term mean (Fig. 4b), which might be due to considerable rainfall and flooding during August 2018. Overall, we also see prominent warming during the monsoon season in Kerala, which can be associated with the declining monsoon season precipitation (H.L. Shah and Mishra, 2016a). We find that despite the monsoon season precipitation was slightly lower than the long-term 95th percentile, total runoff exceeded long-term 95th percentile in August 2018 (Fig. 4c). This variation in total runoff can be attributed to multi-day precipitation extremes during August 2018. Consecutive extreme precipitation days can saturate soils and normally produce more runoff than the precipitation days that are far apart.

Finally, we compared extreme precipitation and total runoff conditions of August 2018 with long-term variability (Fig. 5). To do so, we selected 1 and 2-day maximum precipitation and total runoff conditions in August 2018 and compared them with long-term (1951-2017) values. We find that both 1 and 2-day maximum precipitation that occurred in August 2018 in Kerala exceeded significantly the 95th

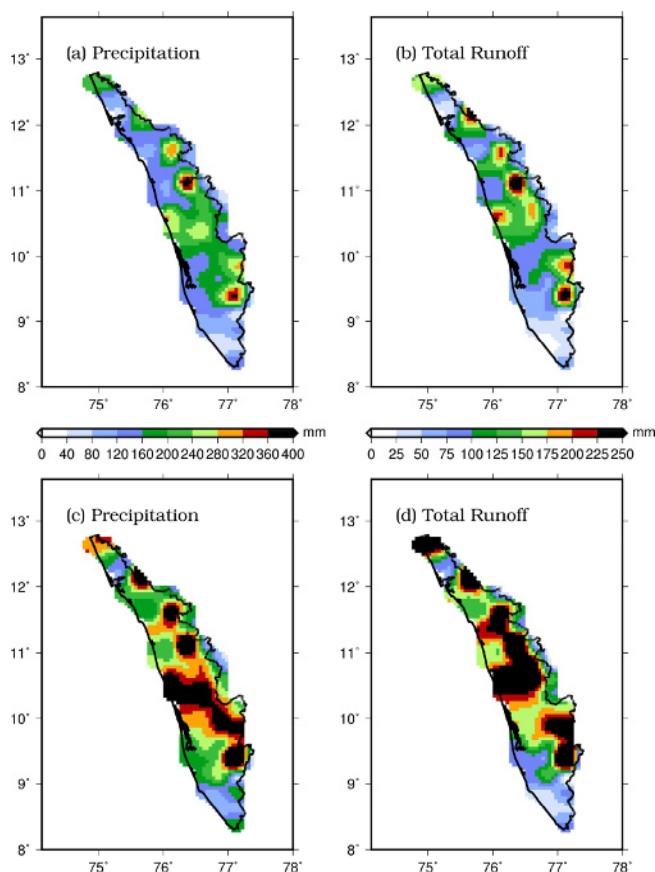


Fig.3. (a-b) 1-day maximum precipitation (mm) and total runoff (mm) during August 2018. **(c-d)** 2-day maximum precipitation (mm) and total runoff (mm) during 2018.

percentile based on 1951-2015 (Fig. 5a, c). For instance, 1-day maximum precipitation that was experienced on 15th August 2018 exceeded by 30mm from 95th percentile of 1-day annual maximum precipitation. Similarly, the 2-day maximum precipitation in August 2018 exceeded its 95th percentile for the 1951-2015 period by 75mm (Fig. 5c). These results show the enormity of extreme precipitation during August 2018. Mishra et al. (2018) reported that 1 and 2-day maximum precipitation in August 2018 had return periods of 75 and 200 years, which indicate the consecutive precipitation occurred on 15 and 16th August might have contributed significantly in the state-wide flooding in Kerala. We noticed similar variations in 1 and 2-day maximum total runoff, which were the highest during the entire record of 1951 and 2018 (Fig. 5b, d). Extreme total runoff in August 2018 was well exceeded its long-term 95th percentile during 1951-2015. Overall, we find that both extreme precipitation and total runoff in August 2018 in Kerala were unprecedented during the past record of 1951-2017, which contributed substantially in the state-wide flooding.

DISCUSSION AND CONCLUSIONS

We provide a long-term hydroclimatic perspective on the conditions that caused the Kerala flooding during August 2018. We find that the hydroclimatic conditions that occurred in August 2018 caused flooding in Kerala are in contrast with the long-term trends. For instance, our results show that Kerala experienced a significant decline in the monsoon season precipitation during the period of 1951-2017. Moreover, the state has witnessed a significant rise in the mean temperature during the monsoon season over the same period. Therefore, long-term trends indicate drying and warming in the monsoon season in Kerala. We find that mean monsoon season total runoff has also declined significantly during 1951-2017. Trends in

extreme hydroclimatic condition in Kerala are consistent with those observed in the mean climate. For instance, both 1 and 2-day annual maximum precipitation and total runoff have declined during the observed record of 1951-2017. These changes in hydroclimatic conditions raise some questions on unprecedented extreme precipitation and runoff conditions that caused flooding in Kerala in August 2018.

The first question that arises from our analysis is if the August 2018 event was caused by climate change or not. Since we do not find any increase in the mean or extreme monsoon season precipitation during 1951-2017, attributing the 2018 flood in Kerala to climate change can be difficult. For instance, despite the significant warming observed in the last 66 years in Kerala mean and extreme precipitation and total runoff have not increased. Our results highlight that the 2018 extreme precipitation and flood events might have caused due to

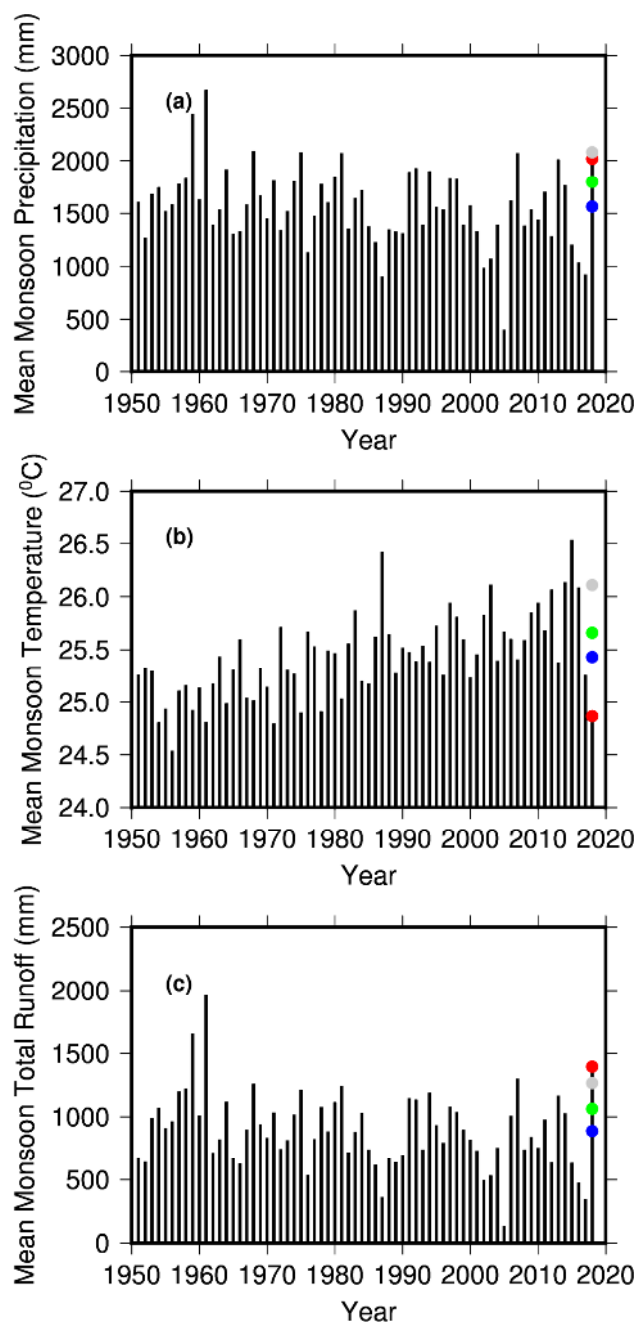


Fig.4. (a-c) Mean monsoon precipitation, temperature and total runoff from 1951-2018 over Kerala. The red circle indicates magnitude during the monsoon season of 2018, which gray, green, and blue indicate long-term 95th, 75th percentiles, and mean, respectively.

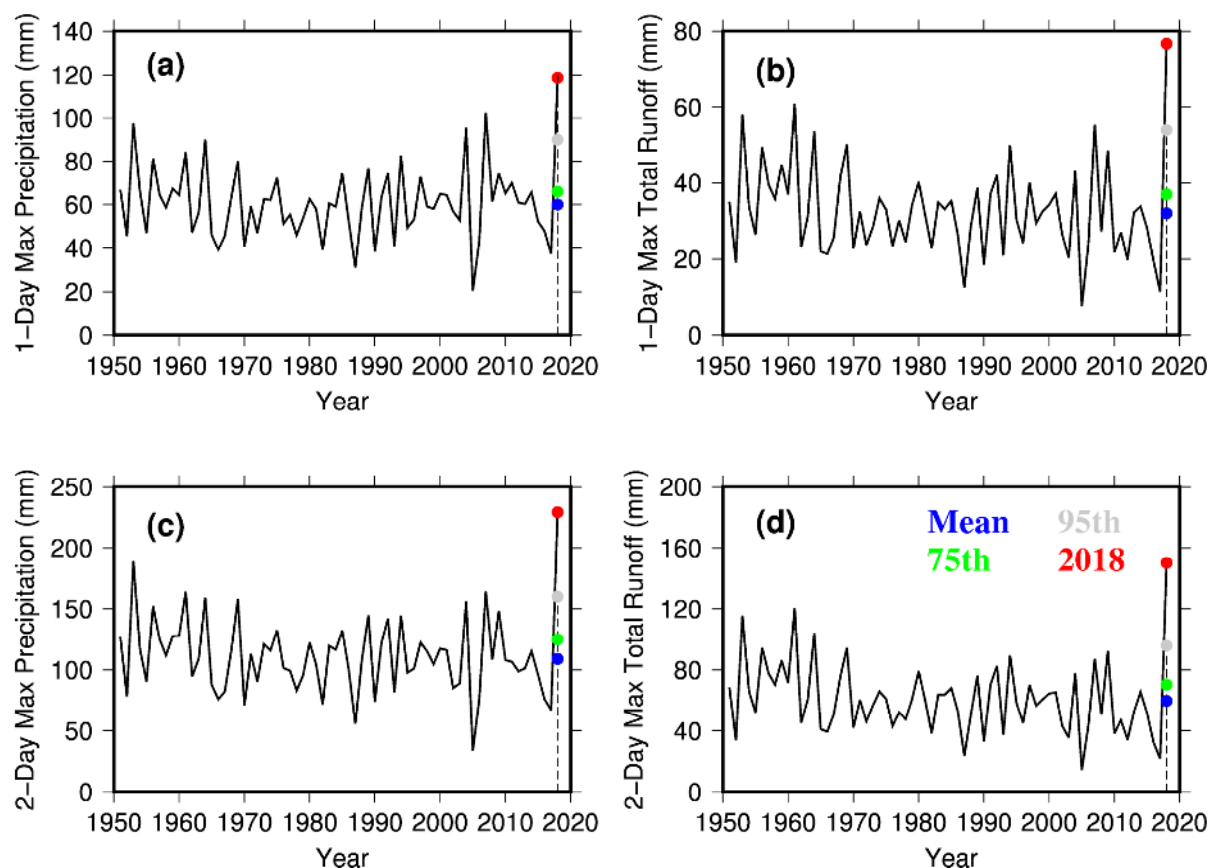


Fig.5. (a-b) Kerala average 1-day maximum precipitation (mm) and total runoff (mm) from 1951- 2018. **(c-d)** Kerala average 2-day maximum precipitation (mm) and total runoff (mm) from 1951- 2018. The red circle indicates magnitude during the monsoon season of 2018, which gray, green, and blue indicate long-term 95th, 75th percentiles, and mean, respectively.

anomalous large-scale circulation features. However, in the future, further research is required to find out the causes of anomalous features that resulted in extreme precipitation in August 2018 in Kerala.

The second important question is—if the flooding in Kerala was caused by the anthropogenic factors other than climate change. Previous studies (Paul et al., 2018) have highlighted the role of deforestation on the monsoon season precipitation in the Western Ghats. Significant changes in land use and land cover affect the land surface hydrologic conditions (Frans et al., 2013; Mishra et al., 2010), which in turn can affect the flood peaks and inundated area. However, how much impact the historic land cover change in Kerala caused on the severity of flood in August 2018 is yet to be understood. The role of reservoirs might have played a key role in at least worsening the flood situation in the state during August 2018 (Mishra et al., 2018). For instance, Mishra et al. (2018) reported that above seasonal rainfall caused above normal reservoir storage prior to the extreme precipitation (15-16 August) in Kerala. The extreme precipitation in the catchments of major reservoirs forced reservoir operators to open the gates and release a considerable amount of stored water, which might have added to the severity of floods in the state.

The last and probably the most important question is—what has to be done to reduce the severity and damage of such floods in the future? The flood of 2018 in Kerala and other floods that caused enormous damage in different parts of the country can provide important insights. For instance, we need a considerable amount of future research to understand the natural and anthropogenic factors associated to flood generation process. For instance, the linkage between land surface conditions (land use land cover, soil, and topography) and floods need to be established in different basins in India, in particular, those that get frequent flooding. Moreover, the

relationship between land surface initial hydrological conditions and floods need to be properly understood. The impacts of climate variability and climate change on the flood frequency in different parts of the country warrants a careful examination. The flood plain delineation for different flood magnitudes, especially with large return periods, is required to minimize the damage to infrastructure and loss of lives. The other important aspect is linked with the extended range forecast of extreme precipitation that can enable hydrologic community and reservoir managers to make proper decisions related to reservoir operations. Managing water resources for agriculture and protecting from floods remains a profound challenge for reservoir operations in the country.

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